Outline of the Course

1. The Learning Problem (*April 3*)
2. Is Learning Feasible? (*April 5*)
3. The Linear Model I (*April 10*)
4. Error and Noise (*April 12*)
5. Training versus Testing (*April 17*)
6. Theory of Generalization (*April 19*)
7. The VC Dimension (*April 24*)
8. Bias-Variance Tradeoff (*April 26*)
9. The Linear Model II (*May 1*)
10. Neural Networks (*May 3*)
11. Overfitting (*May 8*)
12. Regularization (*May 10*)
13. Validation (*May 15*)
14. Support Vector Machines (*May 17*)
15. Kernel Methods (*May 22*)
16. Radial Basis Functions (*May 24*)
17. Three Learning Principles (*May 29*)
18. Epilogue (*May 31*)

- theory; mathematical
- technique; practical
- analysis; conceptual

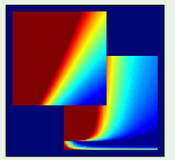
Learning From Data

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Lecture 1: The Learning Problem



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The learning problem - Outline

- Example of machine learning
- Components of Learning
- A simple model
- Types of learning
- Puzzle

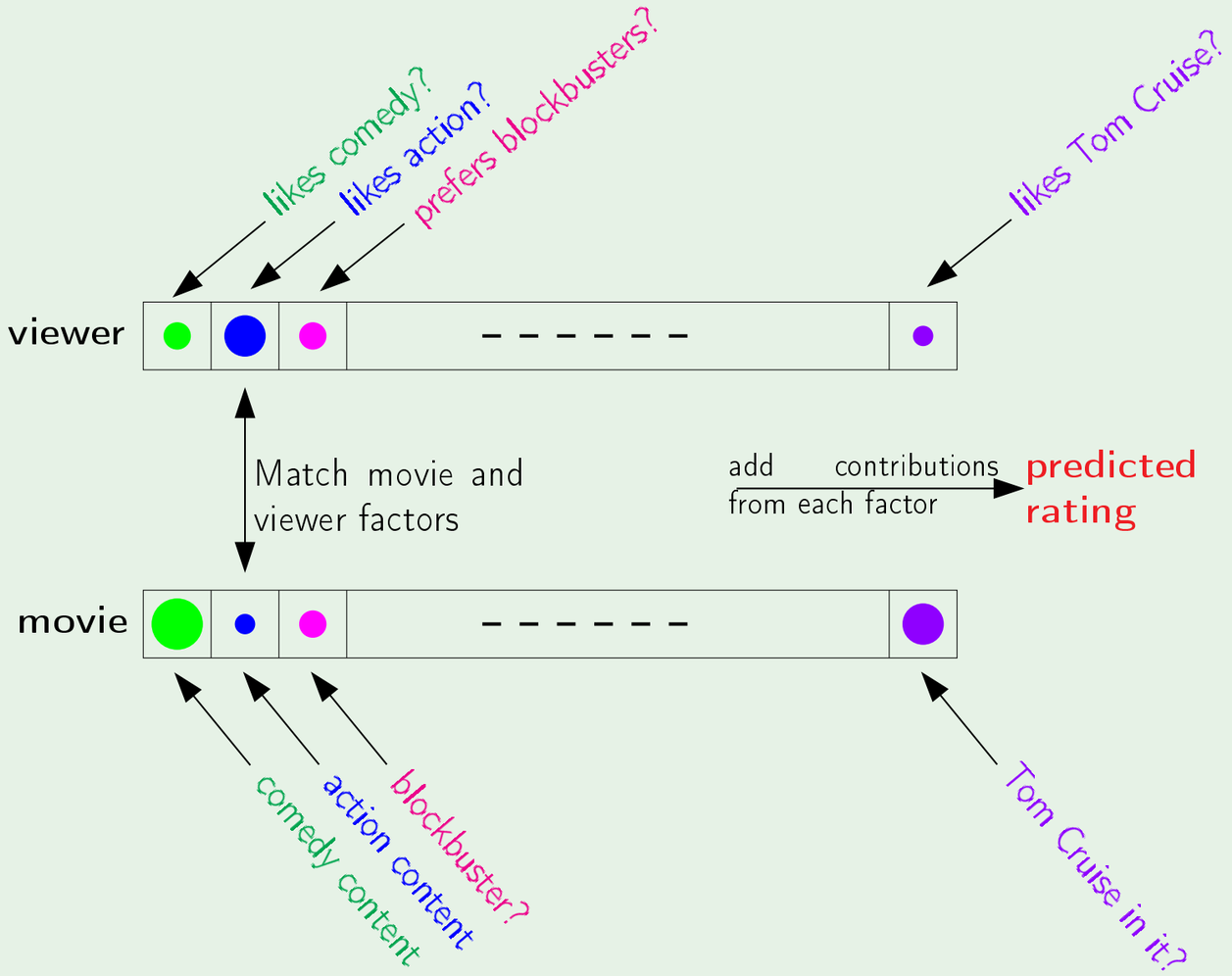
Example: Predicting how a viewer will rate a movie

10% improvement = 1 million dollar prize

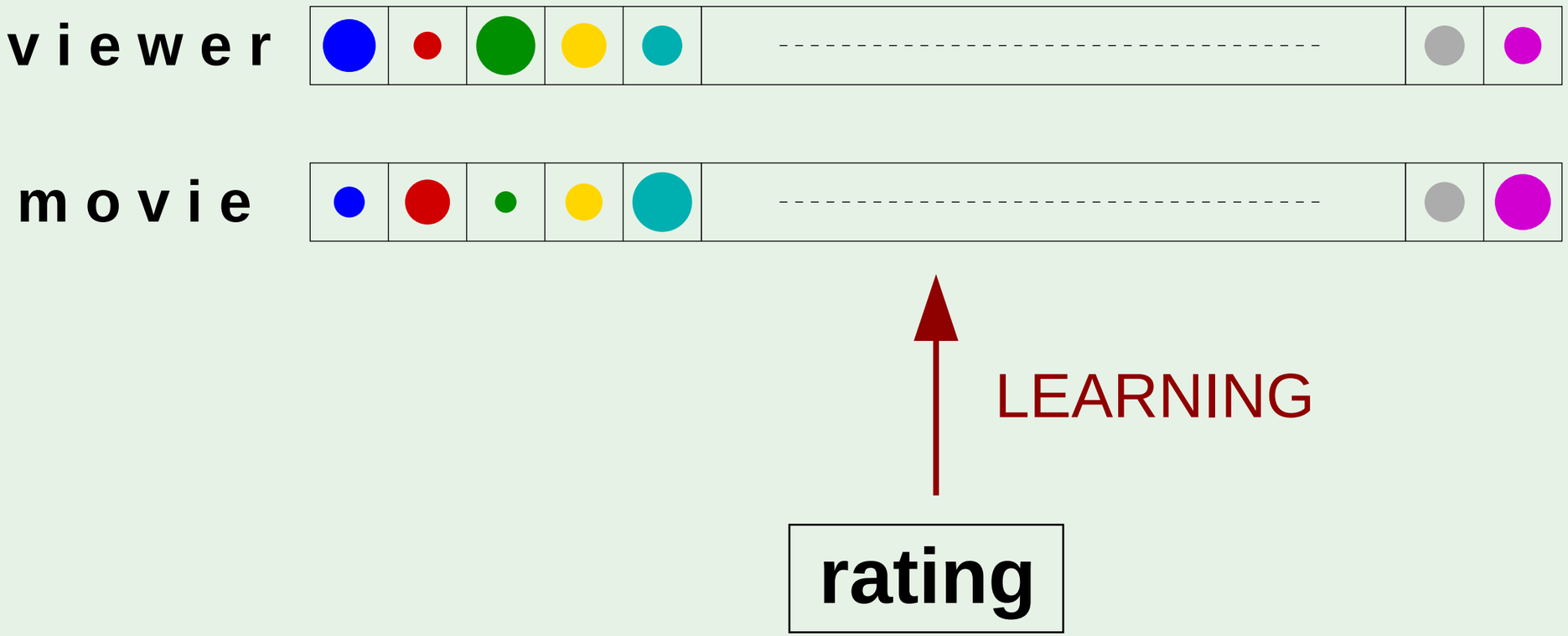
The essence of machine learning:

- A pattern exists.
- We cannot pin it down mathematically.
- We have data on it.

Movie rating - a solution



The learning approach



Components of learning

Metaphor: Credit approval

Applicant information:

age	23 years
gender	male
annual salary	$30,000
years in residence	1 year
years in job	1 year
current debt	$15,000
...	...

Approve credit?

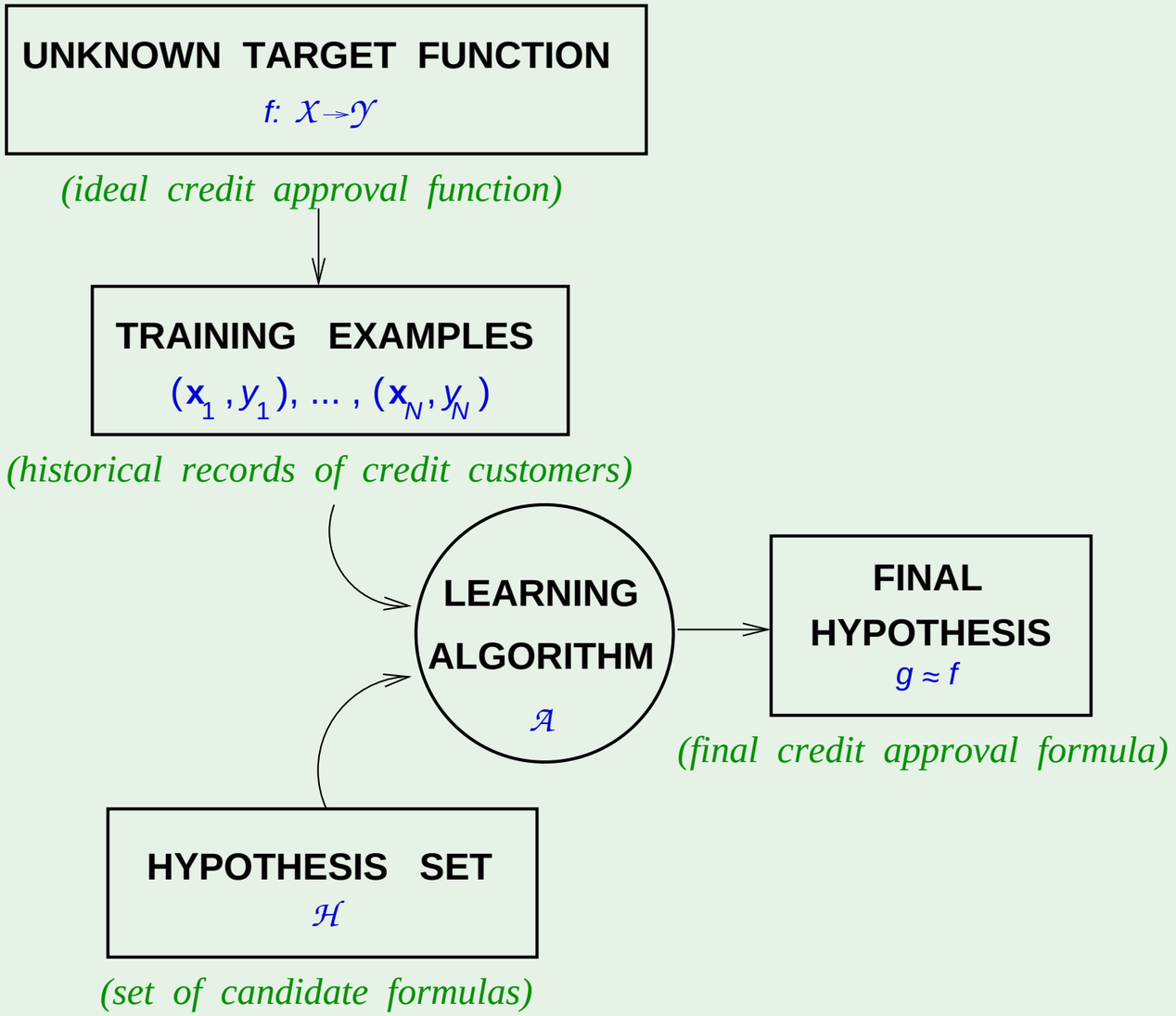
Components of learning

Formalization:

- Input: $\mathbf{x}$ (*customer application*)
- Output: y (*good/bad customer?*)
- Target function: $f : \mathcal{X} \to \mathcal{Y}$ (*ideal credit approval formula*)
- Data: $(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \cdots, (\mathbf{x}_N, y_N)$ (*historical records*)

$\downarrow \quad \downarrow \quad \downarrow$

- Hypothesis: $g : \mathcal{X} \to \mathcal{Y}$ (*formula to be used*)


UNKNOWN TARGET FUNCTION
f: X→Y
(ideal credit approval function)
TRAINING EXAMPLES
(x1 ,y1), ... , (xN ,yN)
(historical records of credit customers)
LEARNING
ALGORITHM
A
FINAL
HYPOTHESIS
g ≈ f
(final credit approval formula)
HYPOTHESIS SET
H
(set of candidate formulas)

Solution components

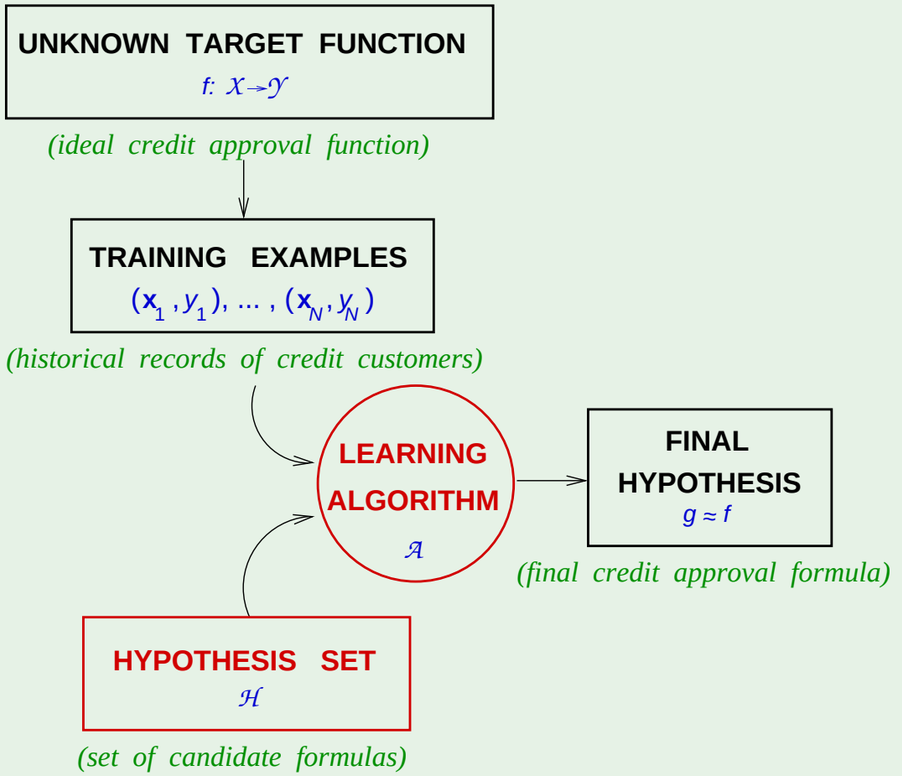
The 2 solution components of the learning problem:

- The Hypothesis Set

  $\mathcal{H} = \{h\}$ $\quad$ $g \in \mathcal{H}$

- The Learning Algorithm

Together, they are referred to as the *learning model*.



A simple hypothesis set - the 'perceptron'

For input $\mathbf{x} = (x_1, \cdots, x_d)$ 'attributes of a customer'

$$\text{Approve credit if} \quad \sum_{i=1}^{d} w_i x_i > \text{threshold},$$

$$\text{Deny credit if} \quad \sum_{i=1}^{d} w_i x_i < \text{threshold}.$$

This linear formula $h \in \mathcal{H}$ can be written as

$$h(\mathbf{x}) = \text{sign}\left(\left(\sum_{i=1}^{d} w_i x_i\right) - \text{threshold}\right)$$

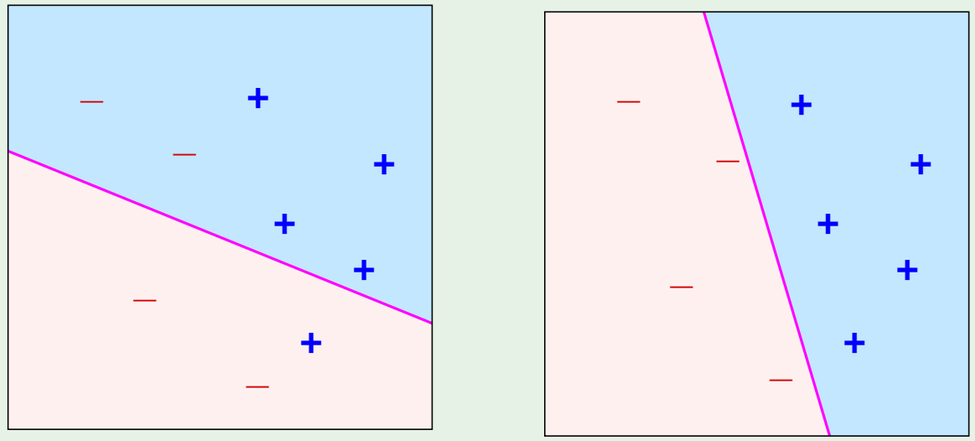
$$h(\mathbf{x}) = \text{sign}\left(\left(\sum_{i=1}^{d} w_i\, x_i\right) + \quad w_0 \quad\right)$$

Introduce an artificial coordinate $x_0 = 1$:

$$h(\mathbf{x}) = \text{sign}\left(\sum_{i=0}^{d} w_i\, x_i\right)$$

In vector form, the perceptron implements

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x})$$



'linearly separable' data

A simple learning algorithm - PLA

The perceptron implements

$$h(\mathbf{x}) = \text{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x})$$

Given the training set:

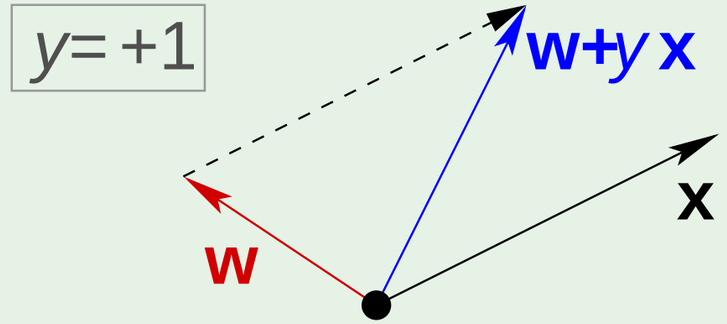
$$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \cdots, (\mathbf{x}_N, y_N)$$

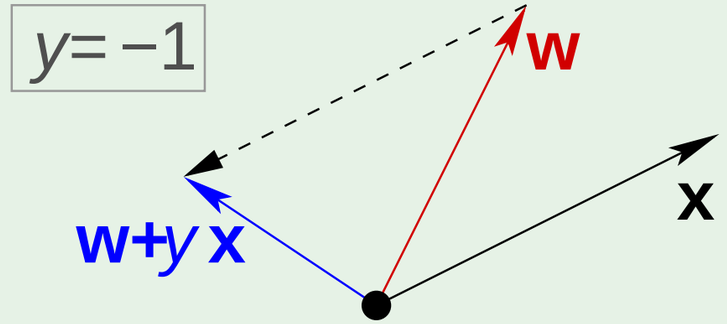
pick a misclassified point:

$$\text{sign}(\mathbf{w}^{\mathsf{T}}\mathbf{x}_n) \neq y_n$$

and update the weight vector:

$$\boxed{\mathbf{w} \leftarrow \mathbf{w} + y_n\mathbf{x}_n}$$





Iterations of PLA

- One iteration of the PLA:

$$\mathbf{w} \leftarrow \mathbf{w} + y\mathbf{x}$$

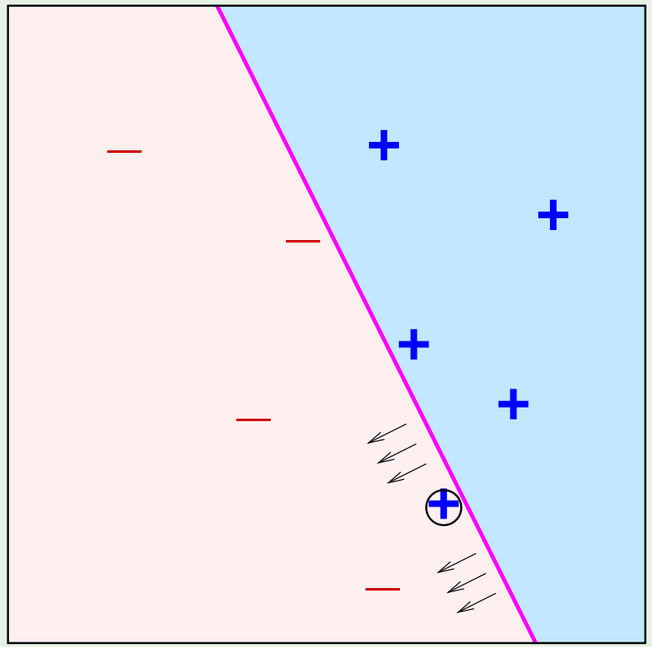
where $(\mathbf{x}, y)$ is a misclassified training point.

- At iteration $t = 1, 2, 3, \cdots$, pick a misclassified point from

$$(\mathbf{x}_1, y_1), (\mathbf{x}_2, y_2), \cdots, (\mathbf{x}_N, y_N)$$

and run a PLA iteration on it.

- That's it!



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Basic premise of learning

"using a set of observations to uncover an underlying process"

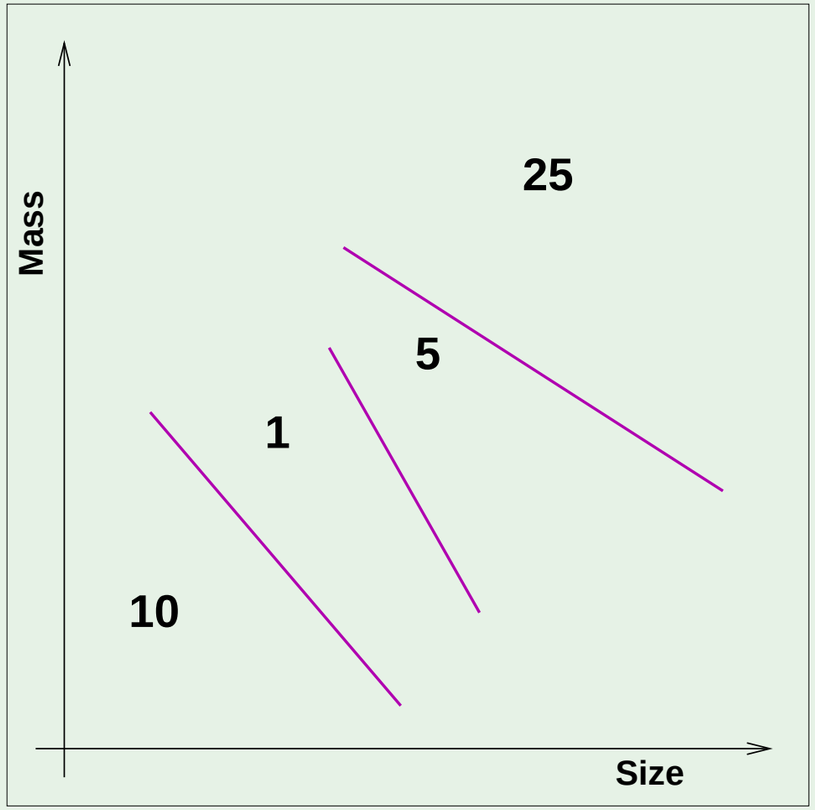
broad premise $\Longrightarrow$ many variations

- Supervised Learning
- Unsupervised Learning
- Reinforcement Learning

Supervised learning

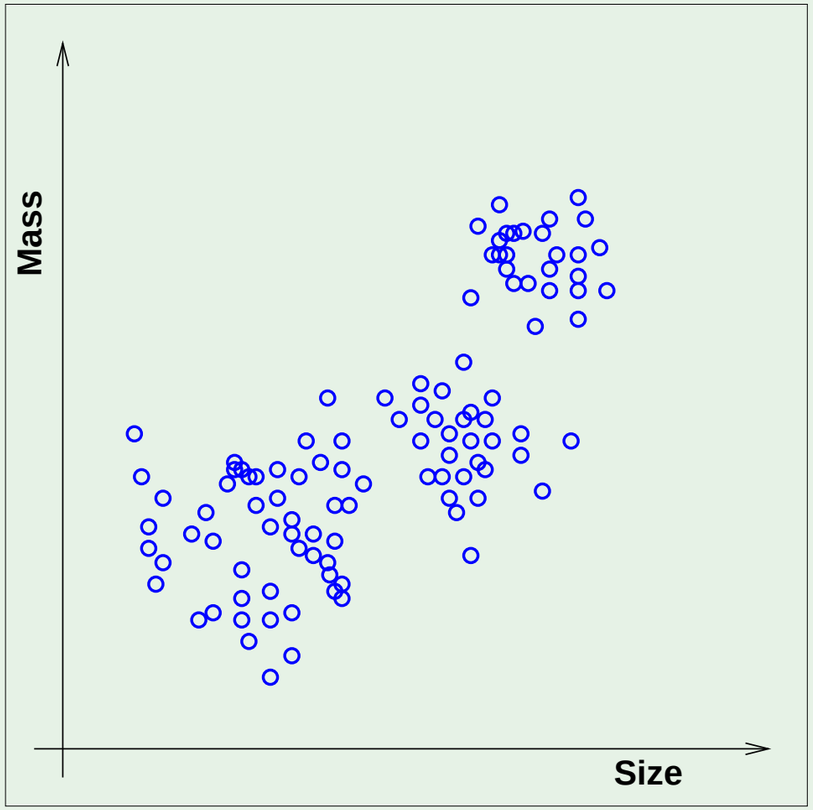
Example from vending machines – **coin recognition**





Unsupervised learning

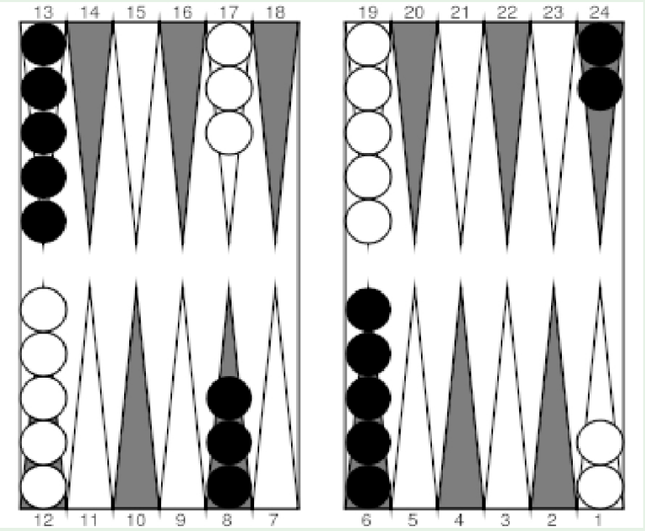
Instead of (input,correct output), we get (input, ?)



Reinforcement learning

Instead of (input,correct output),

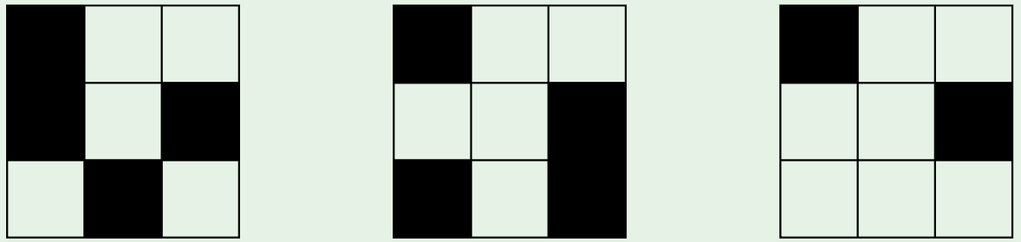
we get (input,*some* output,grade for this output)



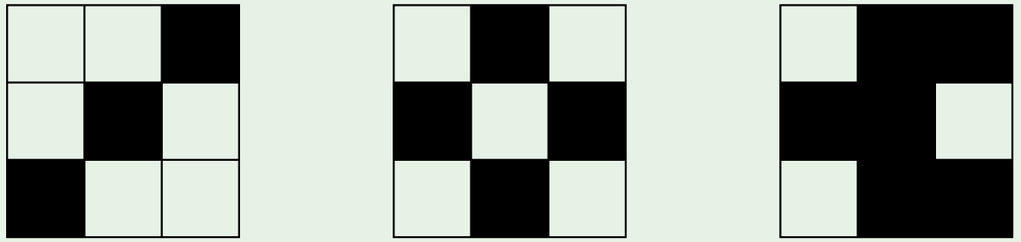


The world champion was a neural network!

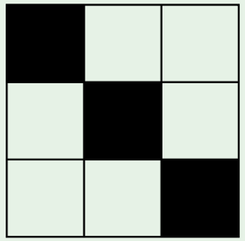
A Learning puzzle



$f = -1$



$f = +1$



$f = ?$